

A TALE OF TWO DENSITIES: ADJACENT GIANT BREAST MASSES WITH DISCORDANT MAMMOGRAPHIC FEATURES

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CASE PRESENTATION

- 61-year-old woman
- Right breast lump since 2014 with rapid enlargement over 6 months
- Previous right breast surgery in 2009 - HPE fibroadenoma
- Examination noted large palpable right breast mass with no overlying skin changes or nipple retraction. No palpable axillary lymph nodes.
- Clinical concern: malignancy

DIAGNOSTIC DILEMMA

Two separate lesions or one heterogeneous giant mass?

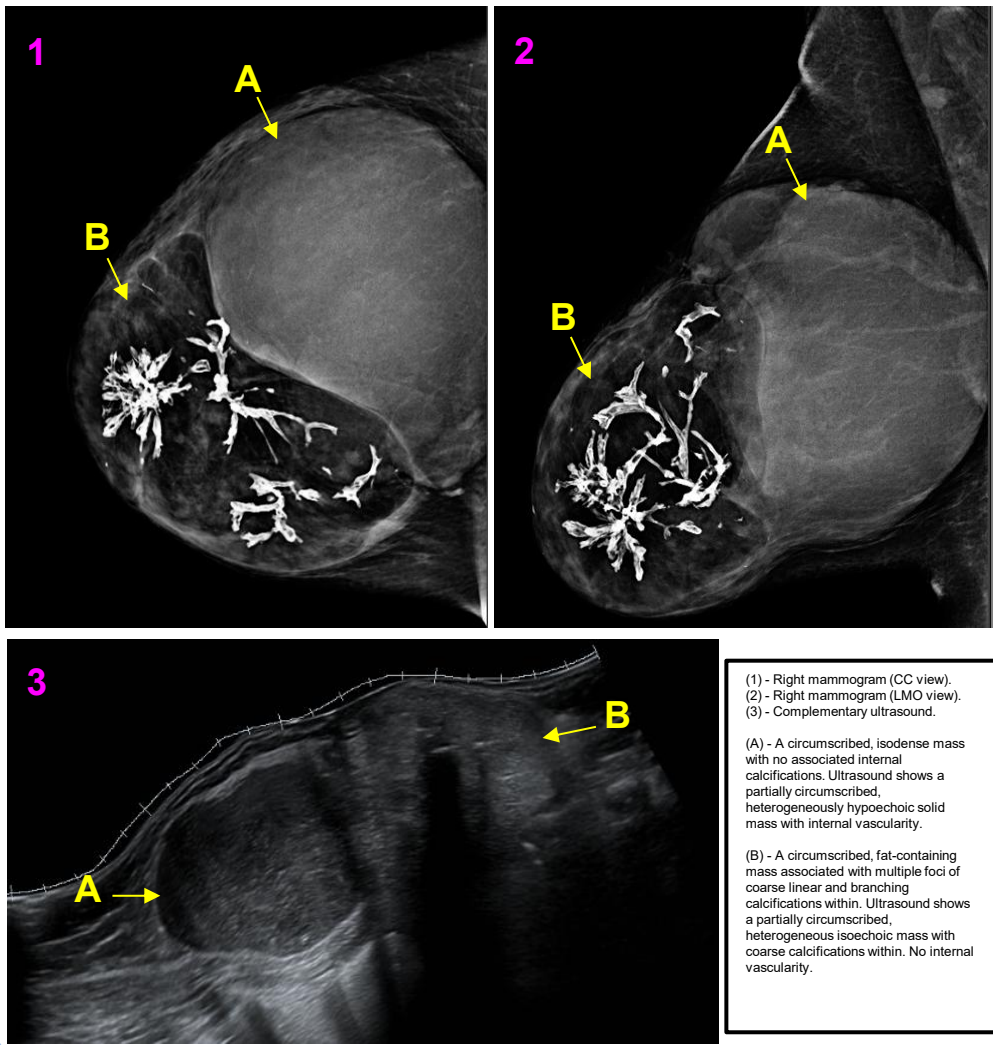
Rapid growth, large size and fibroepithelial features raised concern for giant fibroadenoma, phyllodes tumour or other fibroepithelial lesions.

Meanwhile, the isoechoic mass with coarse calcifications could represent breast hamartoma or giant postsurgical fat necrosis.

BI-RADS 4A

Targeted biopsy of each distinct component was performed.

IMAGING FINDINGS



(1) - Right mammogram (CC view).
(2) - Right mammogram (LMO view).
(3) - Complementary ultrasound.

(A) - A circumscribed, isodense mass with no associated internal calcifications. Ultrasound shows a partially circumscribed, heterogeneously hypoechoic solid mass with internal vascularity.

(B) - A circumscribed, fat-containing mass associated with multiple foci of coarse linear and branching calcifications within. Ultrasound shows a partially circumscribed, heterogeneous isoechoic mass with coarse calcifications within. No internal vascularity.

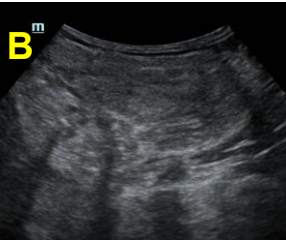
HISTOPATHOLOGY



Soft-tissue density component

HPE: Benign fibroepithelial lesion. No atypia or malignancy.

Interpretation: Benign fibroepithelial process.



Fat-containing calcified component

HPE: Fibrofatty tissue with focal dense hyalinization. No malignancy.

Interpretation: Chronic post-surgical fat necrosis is favoured. (Based on previous surgery, macroscopic fat, dystrophic calcifications and fibrofatty histology.)

LEARNING POINTS

- Large breast masses may contain radiologically distinct components with different pathological processes.
- Rapid growth and giant size may raise suspicion for phyllodes tumour, but benign mimics should also be considered.
- Chronic post-surgical fat necrosis can present as a large mass with coarse calcifications and mimic malignancy.
- Targeted sampling of different imaging components is essential for accurate radiological-pathological correlation.